

MAAZ ALI MATTEHALLI

Innovation and Product Strategy Engineer.

Goa, India | +91 7776971531 | maazali059@gmail.com | [linkedin.com/in/maazali11](https://www.linkedin.com/in/maazali11) | [Portfolio: maazali059.github.io/portfolio](https://portfolio.maazali059.github.io/portfolio)

SUMMARY

Product-minded engineer (B.Tech EEE, NIT Goa, 2026) who bridges hardware and software turning messy, real-world problems into structured frameworks, specs, and roadmaps. Independent work spans marketplace unit economics (Urban Company), EV infrastructure reliability and feasibility (Statiq, EV Goa PPP), agricultural ML (hyperspectral crop yield), and a 0-to-1 IoT hardware MVP (10/10, 75% cost reduction). Professional experience includes contributing research to a national AI-adoption study commissioned by MeitY and launched by the India AI Mission.

CORE COMPETENCIES

Product: Feature Specs, Unit Economics, User Journey Mapping, Roadmap Prioritization Input, Demand-Supply Modeling, Stakeholder Management

Technical Stack: SQL, IoT Telemetry (MQTT, NodeMCU), Embedded Systems, Predictive ML Modeling, TCO/EBITDA/PPP Financial Modeling, Data Visualization

Domain: B2B SaaS & MSME Digitization, On-Demand Marketplaces, EV Charging Infrastructure, Public-Sector Service Delivery, AgriTech ML

PROFESSIONAL EXPERIENCE

Athena Infonomics — Remote

Sep 2025 – Present

Product Intern (Phase 2) | Jan 2026 – Present

- Designed analytical frameworks for the internal product team, translating complex MSME activity-management workflows into actionable technical feature specifications.
- Conducted market research and competitive analysis across 4 industrial sectors to identify customer needs and inform product development decisions.
- Communicated with stakeholders; Industry Experts, executives, developers, designers, and customers to gather requirements and ensure alignment with business objectives.
- Researched and conducted interviews to map value chains across 4 industrial sectors, identifying scalable analytical frameworks for legacy MSMEs and directly contributing to a national AI-adoption study commissioned by MeitY and officially launched by the India AI Mission at the India AI Summit.

Governance & Service Delivery Intern (Phase 1) | Sep 2025 – Dec 2025

- Conducted market, policy, and institutional research to diagnose structural, system-level challenges across public electricity/water utilities and economic regulators in sub-Saharan Africa and India.
- Developed a techno-financial model to define project scope and assess feasibility and ROI for a new water treatment and delivery system.
- Authored specification documents and technical reports for government stakeholders, translating complex data into actionable infrastructure recommendations.
- Engaged with international and government stakeholders to define technical requirements, ensuring end-to-end data accuracy across public-sector engineering specifications.

Saafwater — Goa, India

Jun 2024 – Apr 2025

Product R&D Intern (B.Tech Major Project)

- Identified a market gap during a routine internship that imported water-quality sensors (₹3-6L) were misaligned with Indian municipal needs and expanded the observation into a full B.Tech capstone project.
- Led the 0-to-1 lifecycle of an industrial-grade STP COD-BOD monitoring MVP, configuring raw IoT hardware telemetry for zero-lag, live data sync with an analytics dashboard for field deployment.
- Standardized component selection and consulted on MVP PCB design, achieving a 75% reduction in procurement and prototype costs while validating scalability.
- Formulated ML models on 100+ industrial chemical baseline calibrations, training predictive regression algorithms to flag real-time sensor anomalies and issue proactive maintenance warnings. Awarded 10/10 for industry collaboration and field-ready MVP.

TARGETED PRODUCT PROJECTS

Urban Company (InstaHelp) — Marketplace Unit Economics Teardown

Independent Case Study

- Diagnosed a structural margin collapse: FY25→FY26 operating revenue grew ~35.9% YoY while the business shifted from net profit to a substantial net loss, driven by SLA-constrained fulfillment costs rather than weak demand.
- Quantified the gap: estimated EBITDA loss of ₹447/order against a Net Transaction Value of ~₹148 in Q4 FY26, isolating cost-to-fulfill as the primary driver.
- Identified root causes ~60% reactive demand against rigid 10-15 min SLA windows, and an unstacked 1-worker-1-job labor model where transit time equals task time.
- Architected a 3-part "Efficiency-First" turnaround: Defined Scope Units (standardized work units + AI-visual verification), Demand Batching (30% discount for flexible 3-hour windows), and an "Anchor-Base" society-level worker assignment model.
- Designed a North Star Metric (Contribution Margin per Order) and Survival Metric (Cost per Fulfilled Hour, target 4.5 jobs/hour) with a phased GTM rollout for high-density micro-markets.

Statiq EV Network — Reliability Field Audit & Recovery Roadmap

Independent Field Study, Goa

- Led field testing across Goa's high-density tourist charging corridors, uncovering a 39% telemetry mismatch rate ("ghost availability") and a 16% session-termination failure rate causing connector lock-ins and support escalations.
- Diagnosed root causes as distributed-systems reliability issues spanning charger hardware (OEM variability), network latency, and backend state-reconciliation delays.
- Mapped impact across two personas (interstate road-trippers, EV rental tourists) and surfaced a "dual-app verification loop" undermining single-app trust vs. competitors (Zeon, Tata Power).
- Architected a 3-phase recovery roadmap: Client-Side Confidence Layer (data-freshness indicators), Session Reconciliation Logic (heartbeat-based state sync + auto-timeout), and Safe Termination Fallback (retry → soft unlock → escalation), plus Day-1 validation metrics (App Abandonment Rate, TTFP latency, OEM-level failure distribution).

EV Charging Infrastructure — Strategic Feasibility Analysis (Goa)

Independent Strategic Analysis

- Built a techno-financial model for a 25-station DC fast-charging (50kW) network under a hyper-local Public-Private Partnership structure, incorporating a 40% capital subsidy (Goa EV Policy 2021) and utilization ramping from 7% to 18% over 5 years.
- Projected cumulative profitability by Year 4 with an estimated 18.5% annual ROI (₹2,000,000 CAPEX/station, ₹7.5/kWh commercial tariff); proved hyper-local PPP as the viable alternative to top-down deployment.

Project Management Platforms — Asana/Jira Teardown & PRD

- Conducted a structured teardown of Asana and Jira, identifying UX friction and workflow bottlenecks in team collaboration tools.
- Authored a 0-to-1 PRD proposing role-based access optimizations to improve cross-functional team efficiency.

Smart Irrigation & Yield Prediction — Hyperspectral Crop Analysis

Academic Research, Jan 2025 – Apr 2025

- Trained ML models on 500+ infrared/hyperspectral images, reaching 85%+ accuracy on crop yield prediction and targeting 90%+ on early disease detection.
- Ran exploratory data analysis and spectral feature extraction to map imaging data to ML-ready inputs.

LEADERSHIP & IMPACT

SPIE Student Chapter, NIT Goa

May 2023 – Jan 2025

President

- Revived a dormant chapter (inactive since 2019), securing \$2,400+ in global grant funding and directing 6+ technical workshops for 400+ active users.
- Built inter-institute partnerships with IIT SPIE chapters and the Goa Technology Association, leading two student startups into incubation.

Saavyas (Flagship Fest), NIT Goa

Nov 2023 – Jul 2024

Core Team & Marketing Head

- Automated communication trackers, cutting team delivery lag by 60% and helping scale the event to 5,000+ attendees.

EDUCATION & CERTIFICATIONS

- National Institute of Technology, Goa — B.Tech, Electrical & Electronics Engineering, Class of 2026 (Major Project: 10/10)
- McKinsey & Company — Forward Program (2025) | Aspire Institute — Aspire Leaders Fellow (2026) | Generative AI Mastermind — Practical AI/ML Applications (2025)